NOTES 19: CLT-BASED INFERENCE FOR PROPORTIONS

Stat 120 | Spring 2026

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Name	Type of Variable(s)	Statistic	Parameter
Mean	Quantitative	$\bar{x}$	μ
Proportion	Categorical	$\hat{p}$	p
Standard Deviation	Quantitative	s	σ
Difference in Proportions	2 Categorical (1 Response, 1 Explanatory)	$\hat{p}_1 - \hat{p}_2$	$p_1 - p_2$
Difference in Means	1 Quantitative (Response) 1 Categorical (Explanatory)	$\bar{x}_1 - \bar{x}_2$	$\mu_1 - \mu_2$
Correlation	2 Quantitative	r	ρ
Slope	2 Quantitative	b_1	β_1

CLT: The Central Limit Theorem (CLT) tells us that if the sample size is big enough and the sample is random,

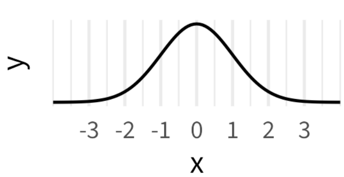
$$\bar{X} \sim N(\underline{\hspace{1.5cm}}, \underline{\hspace{1.5cm}})$$

$$\hat{p} \sim N(\underline{\hspace{1.5cm}}, \underline{\hspace{1.5cm}})$$

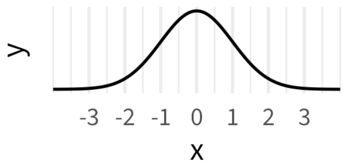
The general form for a **confidence interval** is:

Example: Finding z^*

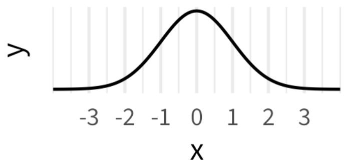
- 95% confidence interval



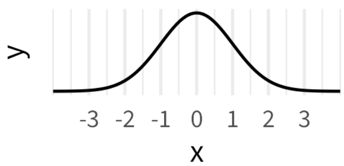
- 68% confidence interval



- 99% confidence interval



- 90% confidence interval



0.1 How big is big enough?

Example 1: $\hat{p} = .5$

Example 1: $\hat{p} = .05$

Rule of Thumb for Proportions:

- Expected count in each category (Yes/No) should be > ____
- $np >$ ____ and $n(1-p) >$ ____

0.2 How do we find the SE?

Standard Error for Proportions:

Idea: As n gets bigger, the SE gets ________________.

If $\hat{p}$ is close to .5, the SE is ________________ than if $\hat{p}$ is close to 0 or 1

Example: ESP Example Again

n = 14

```
p_hat = 3/14
p_0 = .2
n = 14
SE_p = sqrt((p_0*(1-p_0))/n)
SE_p
```

```
[1] 0.1069
```

```
z_score = (p_hat - .2)/SE_p
z_score
```

```
[1] 0.1336
```

```
p_val = pnorm(z_score, lower.tail = FALSE)
p_val
```

```
[1] 0.4468
```

n=1400:

n=14000:

Big Picture Picture

1 In-class Group Questions

1. APM Research Lab ran a survey of likely Minnesota voters between Sept 16-18 of 2024. They reported 48.4% in support of the Harris/Walz ticket, and 43.3% in support of the Trump/Vance ticket.

Here's an excerpt from their methodology report:

> *The margin for error, according to standards customarily used by statisticians, is no more than ± 3.5 percentage points. This means that there is a 95% probability that the "true" figure would fall within that range if all voters were surveyed.*

   (a) Show how the margin of error was computed

   (b) Is it OK to use the normal distribution as a model for the sampling distribution of a proportion in this case?

2. In a random sample of 100 moviegoers in January 2013, 64 of them said they are more likely to wait and watch a new movie in the comfort of their own home.

   (a) Is it ok to use the normal distribution as a model for the sampling distribution of the proportion of moviegoers preferring to wait and watch at home?

   (b) Suppose I want to know if more than 50% of moviegoers in the population prefer to wait and watch at home. Write down the corresponding null and alternative hypotheses.

   (c) For the hypothesis test in part (b), find the standardized test statistic (i.e., the Z-score).

(d) Calculate the p-value and state a conclusion in context

(e) The table below gives some of the percentiles of the N(0,1) distribution. If you were given this table (for example on an exam, where you do not have access to R), what could you say about the value of the p-value? (*Hint:* draw the picture)

percentage	percentile (`qnorm(percentage)`)
90%	1.3
95%	1.6
97.5%	2.0
99%	2.3
99.5%	2.6

(f) Compute and interpret a 90% confidence interval for the population proportion of moviegoers who believe they are more likely to watch a new movie from home. (Use the information from the previous table to find the critical value needed.)

(g) How big of a sample size would you need if you wanted the margin of error for a 90% confidence interval to be no more than 2%? To answer this question, assume that the true proportion is $p = 0.5$, which would result in the biggest margin of error for a given sample size.